Binary S3: checked equivalence and a failed model substitution

Lean proves the ordinary characteristic-two formula S3(b,x,y,z)=(xy+xz+yz)²+xyz+b for y²+xy=x³+a*x²+b. Over an algebraically closed field, on a nonsingular model, vanishing is equivalent to the existence of three actual affine elliptic points with those coordinates and zero sum. No distinct-coordinate assumption remains. [pcc:unverified]

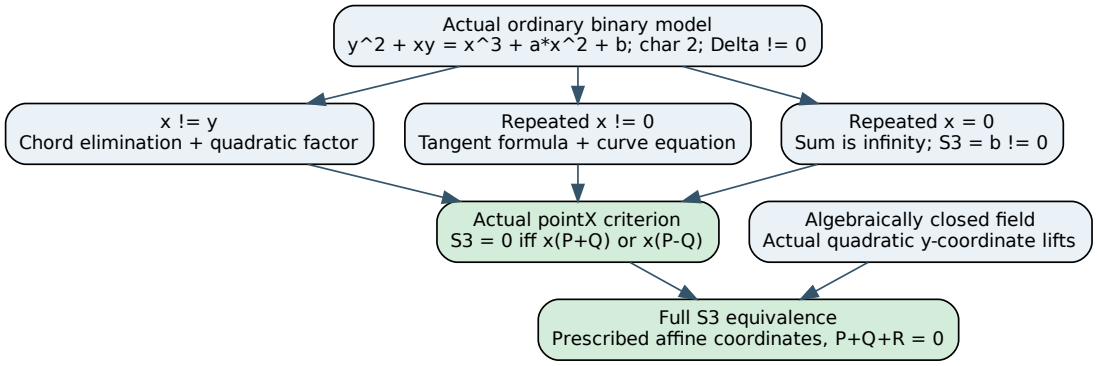

Figure 1: Proof dependencies use actual Mathlib elliptic points, including tangent and infinity cases. The earlier native F2/F4 census is not a premise of this universal theorem.

Case	Checked algebra and group-law behavior
Distinct x	Chord-intercept elimination, two signed-sum roots, quadratic factorization
Repeated nonzero x	Actual tangent; S3=x²(x(2P)−z)
Repeated zero x	Sum at infinity; S3=b≠0, so no finite root
Arbitrary coordinate lifts	Actual degree-two polynomial roots over algebraic closure; Δ=b

The negative control uses the actual nonsingular F2curve y²+xy=x³+1. Lean constructs P,Q,R with x-coordinates1,1,0 and P+Q+R=0. Binary S3 is0; the short-Weierstrass expression is1. This disproves substituting the short expression on a binary model. Neither qualified formula is refuted. [pcc:unverified]

The final whole-project axiom gate passes 116 declarations, with 0 failing and 0 trusted-native exceptions; its receipt is retained beside all failed development attempts. PCC-20261109-517cb7 remains unsigned. No custom axiom, proof hole or trusted native computation is accepted. Main declarations: BinaryS3Audit.binaryS3_zero_iff_exists_points and BinaryS3Audit.short_formula_binary_counterexample. Actual source: ../../lean/BinaryS3.lean and ../../lean/BinaryMisapplication.lean. [pcc:unverified]

This is further item1progress: universal short and ordinary binary S3 equivalences are now formalized. The general resultant recursion and degree2^(m−2) proof remain open. The original Semaev PDF remains an explicitly unread source; this proof independently checks its imported formula. No new native population, clock metric or paid compute. Existing frozen native sources/data/protocols are unchanged. [pcc:unverified]